QBank Quiz December 28, 2024

Test ID: 322522726

Question #1 of 10

Question ID: 1703567

Which is NOT a valid range for private IP addresses?

A) 192.255.255.255 – 193.0.0.0

B) 172.16.0.0 – 172.31.255.255

C) 10.0.0.0 – 10.255.255.255

D) 192.168.0.0 – 192.168.255.255

Explanation

The range 192.255.255.255 – 193.0.0.0 is a valid public IP address range, not a private IP address range.

The Internet Assigned Numbers Authority (IANA) has reserved the following three ranges for private Internet use:

10.0.0.0 – 10.255.255.255 (10.0.0.0/8)
172.16.0.0 – 172.31.255.255 (172.16.0.0/12)
192.168.0.0 – 192.168.255.255 (192.168.0.0/16)

IANA manages and distributes global public IP addresses. IANA also performs DNS root zone management. IANA operates with the help of the International Engineering Task Force (IETF) and RFC Editor to manage IP address allocation and DNS root zone management. There are Regional Internet Registries (RIRs) through which IANA allocates local registrations of IP addresses to different regions of the world. Each RIR handles a specific region of the world.

Objective:
Network Fundamentals

Sub-Objective:
Describe private IPv4 addressing

References:

What Is a Private IP Address?

Question #2 of 10

Question ID: 1704764

The execution of the **show interfaces** command yields the following as a part of its output:

```
Ethernet 0/0 is up, line protocol is down.
```

Which of the following can be determined from this output?

A) The link is not functional due to both a Physical layer and a Data Link layer issue

B) The link is fully functional

C) The link is not functional due to a Physical layer issue

D) **The link is not functional due to a Data Link layer issue**

Explanation

The command output excerpt indicates that the link is not functional due to a Data Link layer (or "line protocol") issue, while the Physical layer (Layer 1) is operational. The first (left) column indicates the Physical layer state of the interface, while the second (right) column indicates the Data Link layer state of the interface.

The link is not fully functional. Were it fully functional, the command output would be:

```
Ethernet0/0 is up, line protocol is up
```

The link is not suffering a Physical layer issue or a combination of Physical layer and Layer 2 (Data Link) layer issues. Were either the case, the output would be:

```
Ethernet 0/0 is down, line protocol is down
```

Note that if a Physical layer issue exists, there will also be a Data Link issue, since the Data Link layer depends on the Physical layer to provide connectivity.

Objective:
Network Fundamentals

Sub-Objective:
Identify interface and cable issues (collisions, errors, mismatch duplex, and/or speed)

References:

Cisco > Product Support > Cisco IOS and NX-OS Software > Cisco IOS Interface and Hardware Component Command Reference > show interfaces

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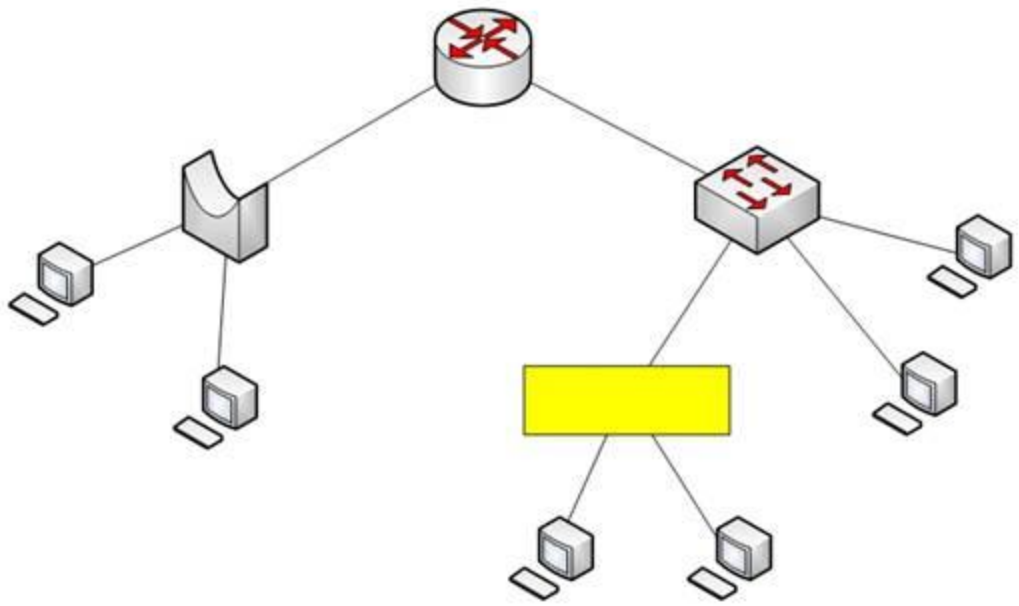
Question #3 of 10

Question ID: 1703510

Assume that all ports on Layer 2 devices are in the same virtual LAN (VLAN). View the given network topology.



Which network device should be placed at the highlighted box to produce a total of two broadcast domains and seven collision domains in the network?

A) Router

B) **Switch**

C) Hub

D) Bridge

Explanation

A hub should be placed at the highlighted box to produce a total of two broadcast domains and seven collision domains in the network. Network devices segment collision domains and broadcast domains in the following manner:

- Hub: A Layer 1 device with all ports in the same collision domain and broadcast domain.
- Bridge/Switch: Layer 2 devices on which all ports are in different collision domains, but in the same broadcast domain (assuming that all ports are in the same VLAN or no VLAN is configured).
- Routers: A Layer 3 device on which every port is a separate collision as well as broadcast domain.

The bridge shown in the graphic has three ports populated by active links, resulting in three collision domains. The switch shown in the exhibit has four ports populated with the links, resulting in four collision domains. Together these two devices create seven collision domains.

Because the scenario requires that there be no more than seven collision domains, the device in the highlighted box must not create any further collision domains. A hub is a device that has all its ports in the same collision domain and will not create any further collision domains in the topology.

A bridge or switch cannot be the correct option because these will also add collision domains.

In the exhibit, the router has two ports with active links, which will result in two broadcast domains. Because the scenario states there are no more than two broadcast domains, the device in the highlighted box must not be a router. Routers are used to segment broadcast domains.

Objective:

Network Fundamentals

Sub-Objective:

Describe characteristics of network topology architectures

References:

Cisco > Product Support > Switches > Catalyst 4500 Series Switch Cisco IOS Software Configuration Guide, 12.2(25)EW > Chapter: Understanding and Configuring VLANs

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Question #4 of 10

Question ID: 1703580

You have an 802.11g wireless network that uses a single wireless access point. For security purposes, you do not broadcast the SSID. You have noticed wireless latency issues on your wireless network.

What should you do to fix this issue?

A) Install another wireless access point that uses a different non-overlapping channel and the same SSID.

B) Install another wireless access point that uses the same non-overlapping channel and a different SSID.

C) **Disable channel bonding.**

D) Enable channel bonding.

Explanation

You would install another wireless access point that uses a different non-overlapping channel and the same SSID. This will allow the connections to be distributed between the two access points, thereby increasing user throughput.

You would not install another wireless access point that uses the same non-overlapping channel and a different SSID. Using the same non-overlapping channel would cause the two access points to have communication issues. Using a different SSID would mean that you would have to physically configure some of the computers to use this new SSID.

You can only use channel bonding on 802.11n wireless networks. While this feature does increase the throughput, thereby reducing latency issues, channel bonding is not available on 802.11g networks.

Objective:

Network Fundamentals

Sub-Objective:

Describe wireless principles

References:

Why channels 1, 6 and 11?

Question #5 of 10

Question ID: 1703572

Which of the following methods of tunneling Internet Protocol version 6 (IPv6) traffic through an IPv4 network increases protocol overhead because of IPv6 headers?

A) IPv6 over dedicated WAN links

B) IPv6 over IPv4 tunnels

C) Dual-stack backbones

D) **Protocol translation**

Explanation

IPv6 over IPv4 tunnels is a method of tunneling IPv6 traffic through an IPv4 network that eliminates the need to create separate circuits to connect to the IPv6 networks. This model increases protocol overhead because of IPv6 headers.

The following deployment models are available for IPv4 to IPv6 migration:

- IPv6 over IPv4 tunnels: IPv6 traffic is encapsulated into IPv4 packets. Then these packets are transferred over IPv4 WAN. This model eliminates the need to create separate circuits to connect to the IPv6 networks. This model increases protocol overhead because of the IPv6 headers and requires both ends to be capable of both protocols.
- Protocol translation: A method allowing an IPv6 host to communicate with an IPv4 host. This is accomplished with the help of Network Address Translation - Protocol Translation (NAT-PT), used to configure translation between IPv6 and IPv4 hosts. NAT-PT allows communication between IPv6 hosts and applications, and native IPv4 hosts and applications.
- IPv6 over dedicated WAN links: A new deployment of IPv6 is created. In this model, IPv6 hierarchy, addressing, and protocols are used by all nodes. However, this model involves cost for creating IPv6 WAN circuits. This solution is not designed for LAN translation but rather translation over WAN links.
- Dual-stack backbones: A hybrid model in which backbone routers have dual-stack functionality, enabling them to route both IPv4 and IPv6 packets. It is suitable for an enterprise that uses both IPv4 and IPv6 applications. Running IPv6 and IPv4 together in a network is known as dual stack routing.

Objective:

Network Fundamentals

Sub-Objective:

Configure and verify IPv6 addressing and prefix

References:

Cisco > Support > Technology Support > IP Version 6 (IPv6) > Configuration Examples and TechNotes > IPv6 Tunnel through an IPv4 Network > Document ID: 25156

Question #6 of 10

Question ID: 1703556

You are the network administrator for your company. You have a Class B address range and are planning for a network that allows 150 hosts per subnet and at least 164 subnets.

Which subnet mask should you use to accomplish the task?

A) 255.255.255.192

B) **255.255.192.0**

C) 255.255.255.252

D) 255.255.255.0

Explanation

You should use 255.255.255.0 as the subnet mask to allow 150 hosts per subnet and at least 164 subnets. The formulas used to calculate the number of subnets and hosts are:

Number of subnets = $2^{\text{number-of-subnet-bits}}$
Number of hosts per subnet = $2^{\text{number-of-host-bits}} - 2$
Subnet mask in decimal: 255.255.255.0
Subnet mask in binary: 11111111.11111111.11111111.00000000
Number of subnet bits: 8 (binary 1s in the subnet octet of the subnet mask)
Number of host bits: 8 (binary 0s in the subnet mask)

In this scenario, we find that for 255.255.255.0:

Subnets that can be used: $2^8 = 256$
Hosts that can be used: $2^8 - 2 = 254$

The other options do not allow 150 hosts per subnet and at least 164 subnets.

If you use 255.255.192.0 as the subnet mask, then the total number of hosts that can be connected per subnet is 16382 (2^{14} - 2 = 16382). However, there will be 4 subnets (2^2 = 4).

If you use 255.255.255.192 as the subnet mask, there will be 62 hosts (2^6 - 2 = 62).

If you use 255.255.255.252 as the subnet mask, there will be two hosts per subnet (2^2 - 2 = 2).

Note: This mask is frequently used for a subnet that connects two routers. In that case, there are two interfaces in the subnet and thus it is the most efficient use of the addressing space. This is also the most efficient way to address a point-to-point serial link.

A note about the formulas: You will always subtract 2 from the number of hosts ($2^{\text{number-of-host-bits}}$ - 2) because the all-zeroes bit address is reserved for the network address and the all-ones bit address is reserved for the broadcast address.

Before Cisco IOS Software Release 12.0, it was common practice to subtract 2 from the networks formula ($2^{\text{number-of-subnet-bits}}$) to exclude the all-ones subnet and subnet zero. Today that range is usable, except with some legacy systems. On certain networks with legacy software, you may need to use the previous formula ($2^{\text{number-of-subnet-bits}}$ - 2) to calculate the number of valid subnets.

Objective:
Network Fundamentals

Sub-Objective:
Configure and verify IPv4 addressing and subnetting

References:

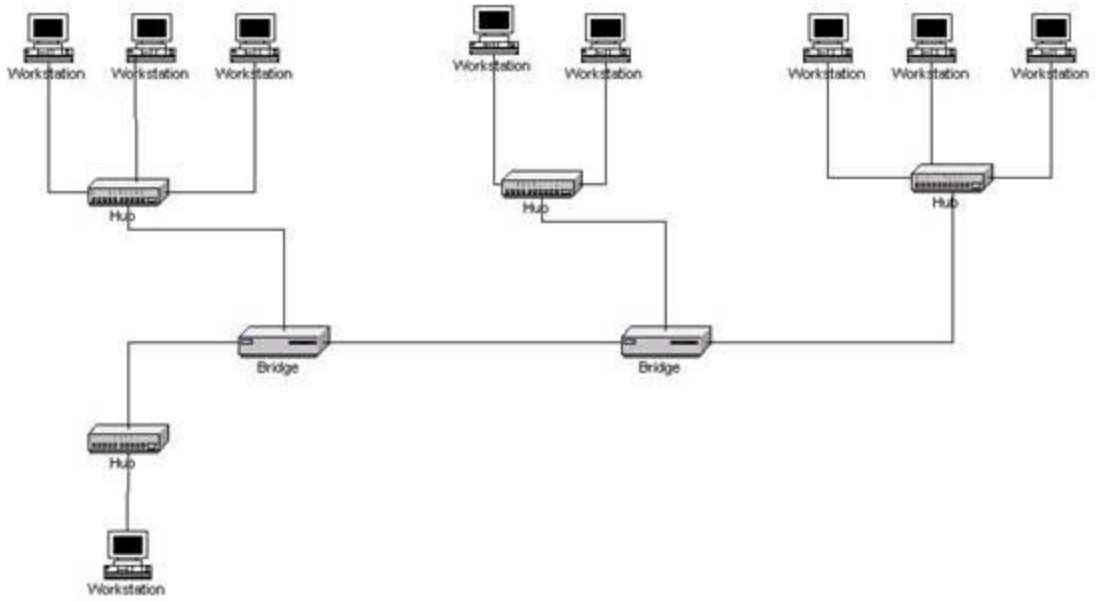
Cisco > Support > Technology Support > IP Routing > Troubleshooting TechNotes > Configure IP Addresses and Unique Subnets for New Users > Document ID: 13788 > Understand IP Addresses

Cisco > Technology Support > IP Addressing Services > Troubleshooting TechNotes > Configure Subnet Zero and All-Ones Subnet > Document ID: 13711

Question #7 of 10

Question ID: 1704747

How many collision domains are in a LAN with four hubs and two bridges that are connected directly to each other, as shown in the following figure?



A) Four

B) Six

C) **Five**

D) Fourteen

Explanation

A bridge segments the LAN into separate collision domains. The figure in this scenario has five segments created between these two bridges. Therefore, there will be five collision domains (segments) on the LAN if the two bridges are directly connected as shown in the exhibit.

Hubs do not create LAN segments; they act as port aggregators and signal amplifiers.

It is also worth noting that, with no router in the diagram, the entire network is a single broadcast domain. If a router were present, each of its interfaces could host a different subnet, and each of those same interfaces would be a separate broadcast domain.

Objective:
Network Fundamentals

Sub-Objective:
Explain the role and function of network components

References:

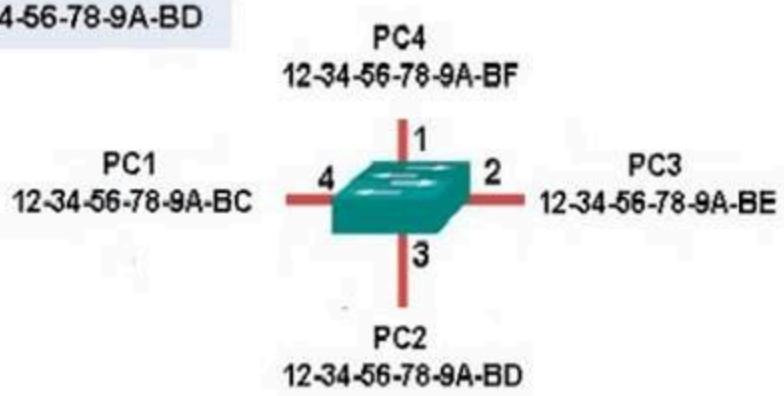
Cisco > Solutions > Small Business IT Solutions > Cisco Small Business Resource Center > Small Business Networking Resources > What is a switch vs. a router?

Question #8 of 10

Question ID: 1583569

The exhibit below displays the MAC address table of a switch in your network, along with the location of each device connected to the switch.

MAC Address Table	
Port	MAC Address
1	12-34-56-78-9A-BF
3	12-34-56-78-9A-BD



Which of the following frames will cause the switch to add a new MAC address to its table and forward the frame to all ports when the frame is received?

A) source MAC: ff–ff–ff–ff–ff, destination MAC: 12–34–56–78–9A–BC

B) source MAC: 12–34–56–78–9A–BC, destination MAC: ff–ff–ff–ff–ff–ff

C) source MAC: 12–34–56–78–9A–BF, destination MAC: 12–34–56–78–9A–BC

D) source MAC: 12–34–56–78–9A–BC, destination MAC: 12–34–56–78–9A–BF

Explanation

The only frame that will be handled in the specified way is the one with a source MAC of 12–34–56–78–9A–BC and a destination MAC of ff–ff–ff–ff–ff–ff. Since the source address 12–34–56–78–9A–BC is not already in the MAC table, the switch will add it. It will forward the frame to all ports because the destination is the broadcast MAC address of ff–ff–ff–ff–ff–ff.

A frame with a source MAC of ff–ff–ff–ff–ff–ff and a destination MAC of 12–34–56–78–9A–BC is an impossible combination. That would mean that the frame is coming from all devices, which is not possible.

The frame with a source MAC of 12–34–56–78–9A–BF and a destination MAC of 12–34–56–78–9A–BC would be sent to all ports because the destination MAC address is not in the MAC address table. However, the switch would not add a new MAC address to the table because the source address is already in the table.

The frame with a source MAC of 12–34–56–78–9A–BC and a destination MAC of 12–34–56–78–9A–BF would not be forwarded to all ports because the destination MAC address is in the table. The switch would add a new MAC address to the table because the source MAC address is not currently in the MAC address table.

Objective:

Network Fundamentals

Sub-Objective:

Describe switching concepts

References:

Cisco Press > Articles > Cisco Certification > CCNA Routing and Switching > Basic Data Transmission in Networks: MAC Tables and ARP Tables

Cisco Press > Articles > Cisco Network Technology > General Networking > Cisco Networking Academy's Introduction to Switched Networks

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Question #9 of 10

Question ID: 1704749

You are the network administrator for your company. The network at the company's office is due to be upgraded, and you have been assigned the responsibility of identifying the requirements for designing the network. You need to provide network connectivity to 200 client computers that will reside in the same sub network, and each client computer must be allocated dedicated bandwidth.

Which device should you use to accomplish the task?

A) Firewall

B) **Switch**

C) Router

D) Hub

Explanation

You would use a switch to accomplish the task in this scenario. A switch is used to provide dedicated bandwidth to each node by eliminating the possibility of collisions on the switch port where the node resides. Switches work at Layer 2 in the Open System Interconnection (OSI) model and perform the function of separating collision domains. When a node resides in its own collision domain, the possibility of collisions (which slow throughput due to the subsequent but necessary retransmission) is eliminated. The advantage of using a switch instead of a hub is that a switch provides dedicated bandwidth to each client, while all connected clients share the bandwidth on a hub.

A router would not be a suitable device in this scenario. Routers are Network layer devices that are used to separate broadcast domains and connect two or more different subnets or network types. There is only a single subnet in the scenario, so a router is not required.

A hub would not be a suitable device in this scenario. Hubs are Physical layer (Layer 1) devices that are used to connect clients to the network. A hub simply broadcasts data to all its ports; it does not create separate collision domains. All clients connected to a hub are a member of a single collision domain. In a scenario where a number of devices connected to a hub are experiencing network slowdowns, especially when using network-based applications, replacing the hub with a switch is almost always the best solution.

A firewall would not be a suitable device in this scenario. A firewall is a device used to secure the network against unauthorized intrusions and malicious attacks.

Objective:
Network Fundamentals

Sub-Objective:
Explain the role and function of network components

References:

Layer 2 vs Layer 3 Network Switches: What's the Difference?

Question #10 of 10

Question ID: 1703543

Which of the following statements are TRUE of a straight-through cable? (Choose two.)

A) You can connect two routers using a straight-through cable.

B) The wires on the cable are crossed over.

C) **You can connect a switch to a router using a straight-through cable.**

D) It is also known as a patch cable.

E) **You can connect a hub to a switch using a straight-through cable.**

Explanation

A straight-through cable is also known as a patch cable, and a straight-through cable is used to connect a switch to a router. The following are the characteristics of a straight-through cable:

- It is a twisted-pair copper wire cable.
- The RJ-45 connectors at both ends have the same conductor arrangement.
- It is also known as a patch cable.

- You can connect a switch to a router using a straight-through cable.
- You can connect a router to a hub or a workstation to a hub using a straight-through cable.

All the other options are incorrect because they are the characteristics of a crossover cable.

Objective:

Network Fundamentals

Sub-Objective:

Identify interface and cable issues (collisions, errors, mismatch duplex, and/or speed)

References:

Cisco > Support > Product Support > Routers > Cisco 7200 Series Routers > Cisco 7200 Series Routers > Design TechNotes > Specifications for Ethernet 100BaseTX and 10BaseT Cables > Document ID: 46792